Customer Segmentation with Clustering

CAM DS C101 Mini Project 6.3.1

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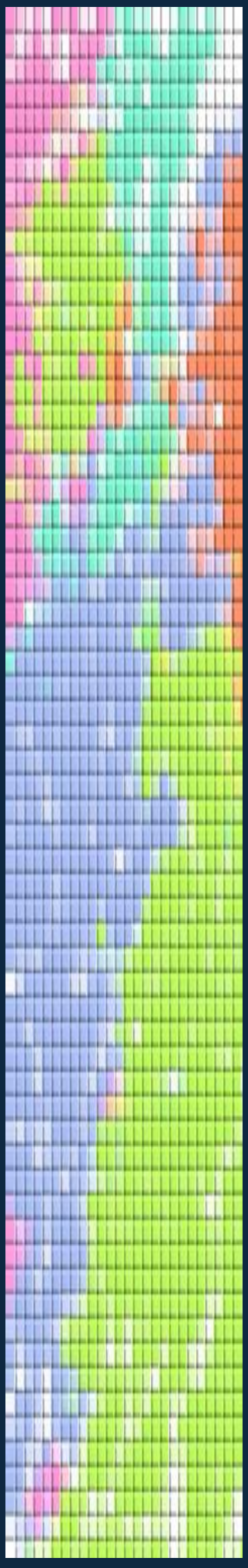


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Introduction

An e-commerce dataset involving a transatlantic customer base has been analysed to perform segmentation using unsupervised machine learning (ML) methods, including *k-means* and *hierarchical* clustering. The dataset contains 951,668 rows and comprises 20 features which convey information regarding orders placed over a 4-year period as well as customer details.

Data Preprocessing & Feature Creation

Following the initial assessment of data for missing values and duplicates, five new features were created based on a marketing perspective as described in Table 1.

Feature Created	Original Feature	Objective
Frequency	Order ID	Captures the frequency of customer activity and informs of the effectiveness of marketing campaigns and how to target less-active customers.
Recency	Delivery Date	Helps predict customer churn (turnover) and engagement.
CLV	Total Revenue	Customer lifetime value (CLV) helps to prioritise marketing efforts and resources as it focuses on customers who are expected to bring the most value over time.
Ave(rage) Unit Cost	Unit Cost	The average unit cost indicates if the customer prefers low-cost or high-cost items and is related to the profitability of purchases.
Customer Age	Customer Birth Date	Relays demographic information about the customer base.

Table 1: The five metrics chosen to infer the nuances of the customer base.

Next, standard scaling was applied to preserve the underlying distributions and minimise sensitivity to potential outliers.

Exploratory Data Analysis

The dataset was further explored using histograms and boxplots as depicted in Figure 1.

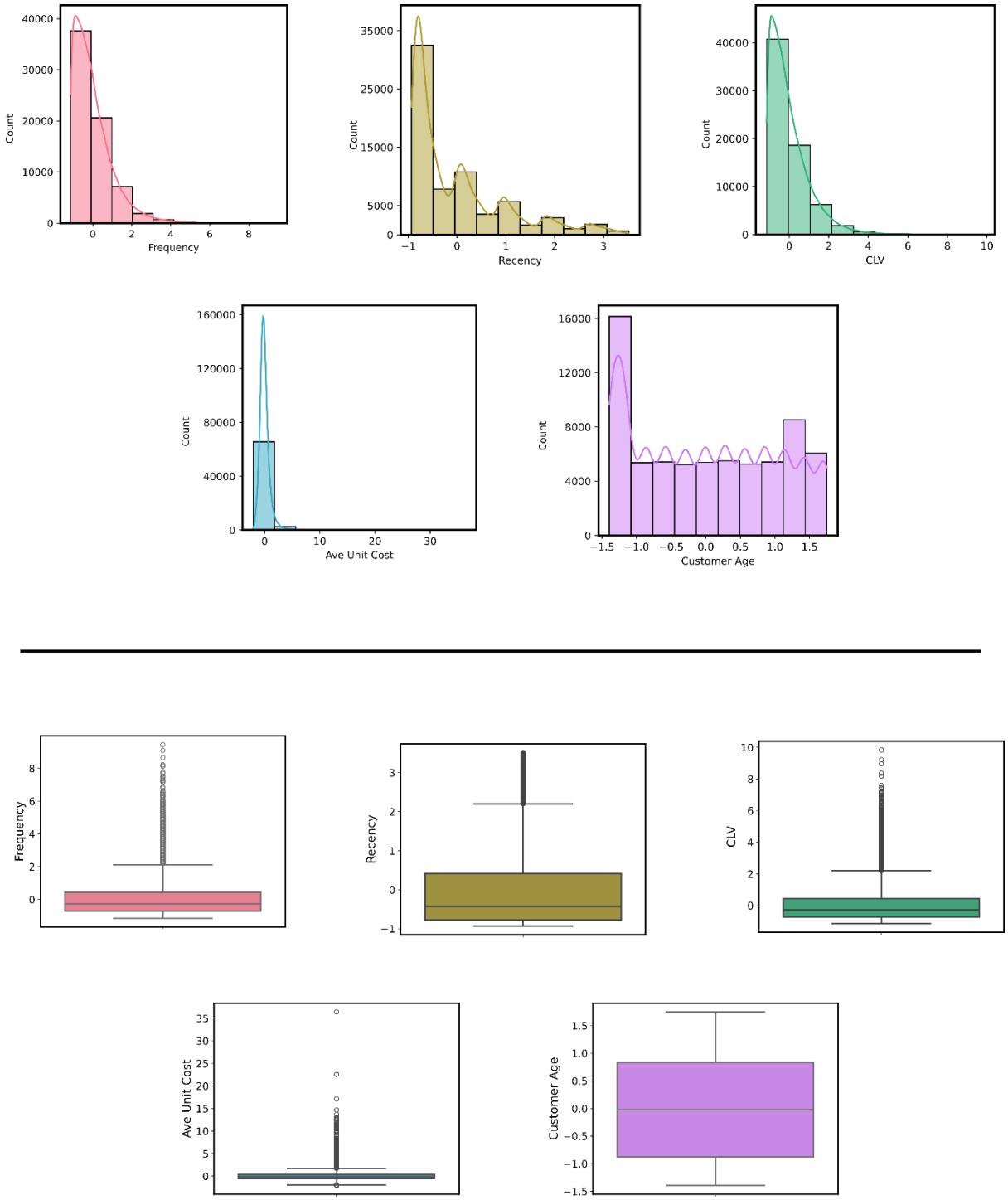


Figure 1: The histograms (top panel) and boxplots (bottom panel), obtained for each feature created.

The histograms show none of the features follow a normal distribution due to a high-level of skewness, whereas the boxplots indicate a high percentage of outliers which are predominantly found above the maximum.

Using a simple analysis based on the interquartile range, the total number of outliers were found to represent 16.82% of the dataset. However, these datapoints could stem from naturally occurring extremes such as customers with unusual purchasing patterns.

Determining the Optimum Number of Clusters

Using K-means Methods

K-means clustering groups data into clusters centred around centroids that are initially randomly assigned. Thus, it is crucial to optimise the number of clusters, k, for which two approaches were implemented:

- The elbow method which plots the *within-cluster sum of squares* (WCSS) against an increasing k, to identify the initial point of decline in the rate of decrease in WCSS.
- The silhouette scores which combine intra-cluster and nearest-cluster distances to score the quality of the clusters formed.

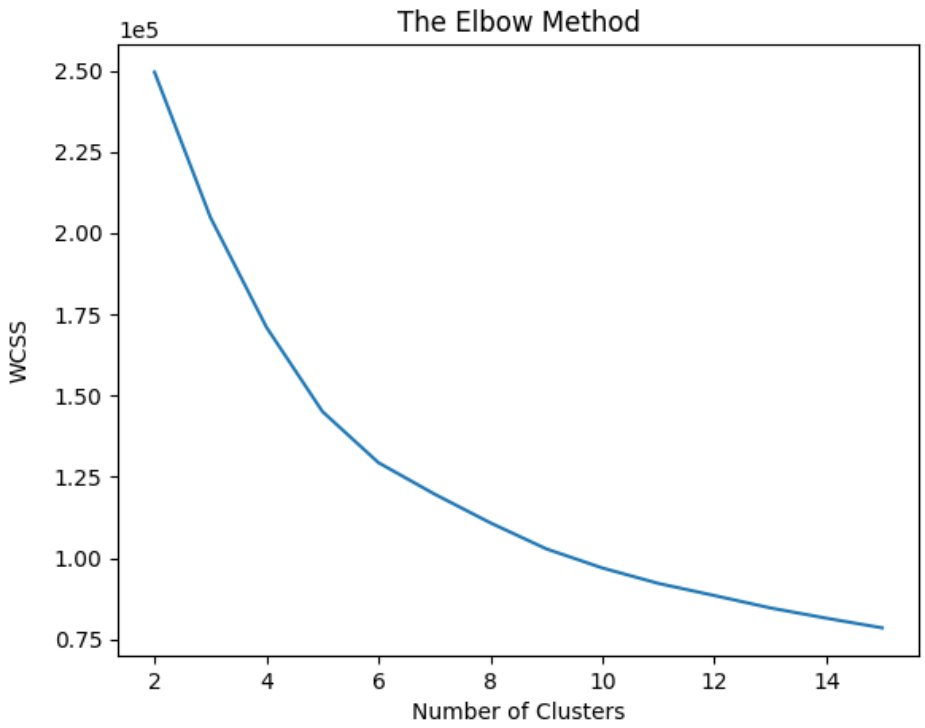


Figure 2: The elbow plot of the in-cluster variation against increasing k.

The elbow plot in Figure 2 suggests the within-cluster spread is decreasing with higher k; but the initial point of plateau is highly subjective, as it can be any point between $\mathrm{k} = 6$

and 8. Based on the silhouette scores presented in Figure 3, the optimal number of clusters is 5. However, the mean score is 0.25 which is closer to 0 than 1, thereby suggesting potential overlaps between neighbouring clusters.

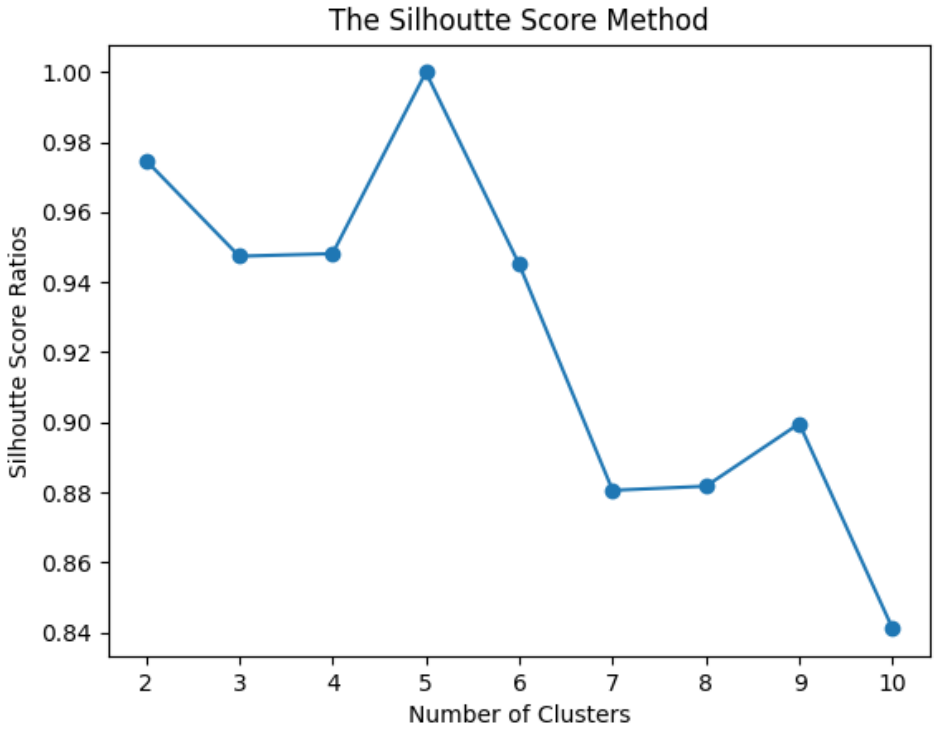


Figure 3: The ratio of silhouette scores to that of the maximum score ($k = 5$), plotted against k. The scores obtained for $k = 3$, 4 and 6 are nearly 95% of the score obtained for $k = 5$.

Using Hierarchical Clustering

A hierarchical structure is formed by grouping similar datapoints into clusters whose distance-relationship is illustrated by a dendrogram. Figure 4 shows a dendrogram produced by performing centroid-linkage on a subset of 30,000 datapoints, where the depth has been pruned to 10 clusters.

Distances ranging from 110 to 150, return 5 clusters with a reasonable separation between the different groups. When the distance is decreased, up to five merges can be eliminated within a relatively small range of 70-100, thereby creating 10 to 6 clusters respectively. Such incremental jumps are non-ideal and paint an obscure picture. Increasing the distance above 150 or 170 yields 4 or 3 clusters respectively, which based on the near-uniform cluster distribution are viable options. However, the case of 5 clusters holds true over a larger distance range while still producing a relatively uniform spread.

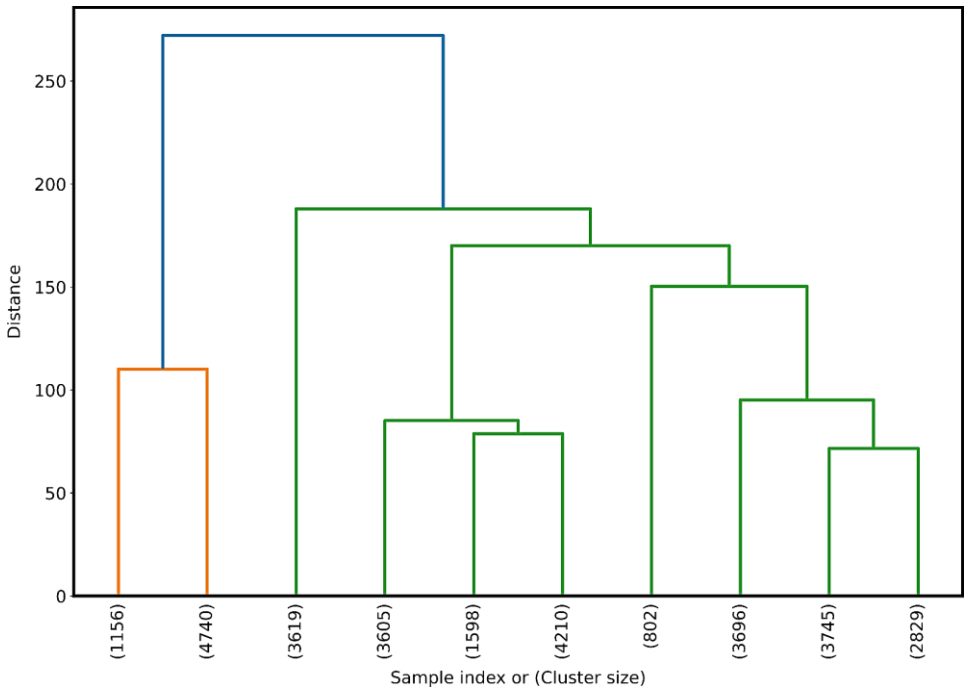


Figure 4: The dendrogram shows a distance-based hierarchical structure formed for the 30,000 datapoints which merge into clusters, starting from the 10 most populated groups.

Performing K-means with Optimal *k*

Based on both the silhouette scores and the dendrogram, an optimal k of 5 was chosen, which is not far from the point of inertia suggested by the elbow plot. Thus, k-means clustering was performed with k =5 and the results obtained are presented as boxplots in Figure 5.

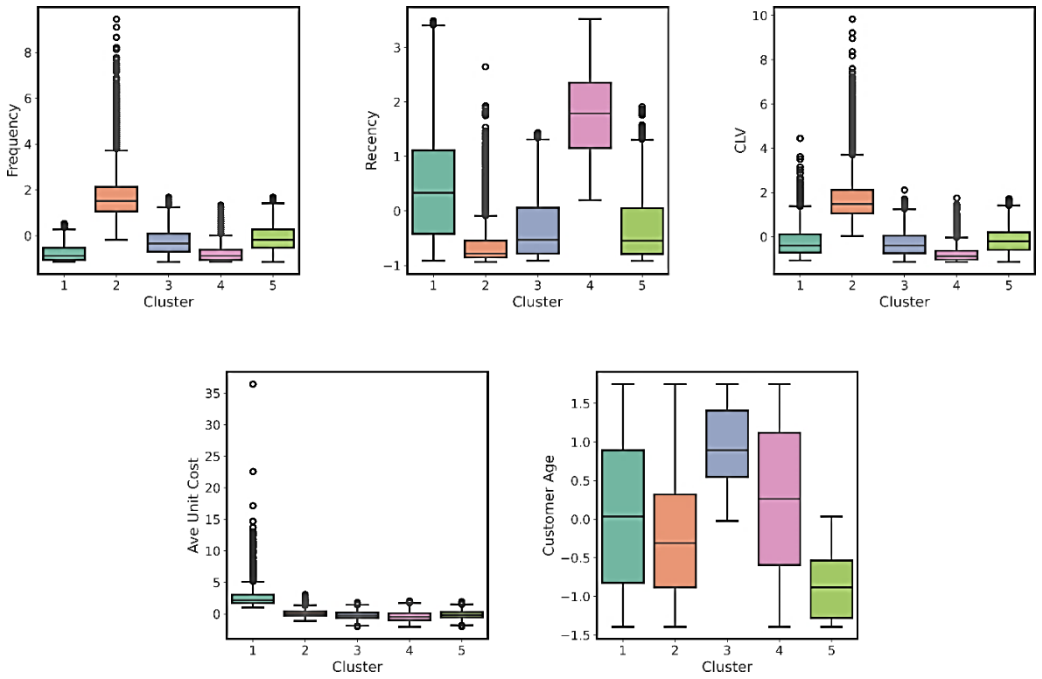


Figure 5: Boxplots showing the distribution of the resulting 5 clusters for frequency, recency, CLV, average unit cost, and customer age.

For each feature, the distribution of the 5 boxes, each representing a cluster, suggests potential overlaps, the extent of which are likely to be exacerbated by the presence of outliers.

2D Visualisation with PCA and t-SNE

Dimensionality reduction of the resulting dataset (assigned with cluster labels) was performed using principal component analysis (PCA) as well as t-distributed stochastic neighbour embedding (t-SNE). The PCA plot in Figure 6 indicates a case of soft clustering where neighbouring clusters overlap significantly. Indeed, cluster 3 appears completely absorbed by the neighbouring clusters. The two principal axes, collectively account for 65.25% of the total variance which indicates considerable loss of structure and interpretability. As a positive, datapoints furthest along the two principal axes are assigned to clusters 1 and 2, which based on the boxplots in Figure 5 are the most populated with outliers.

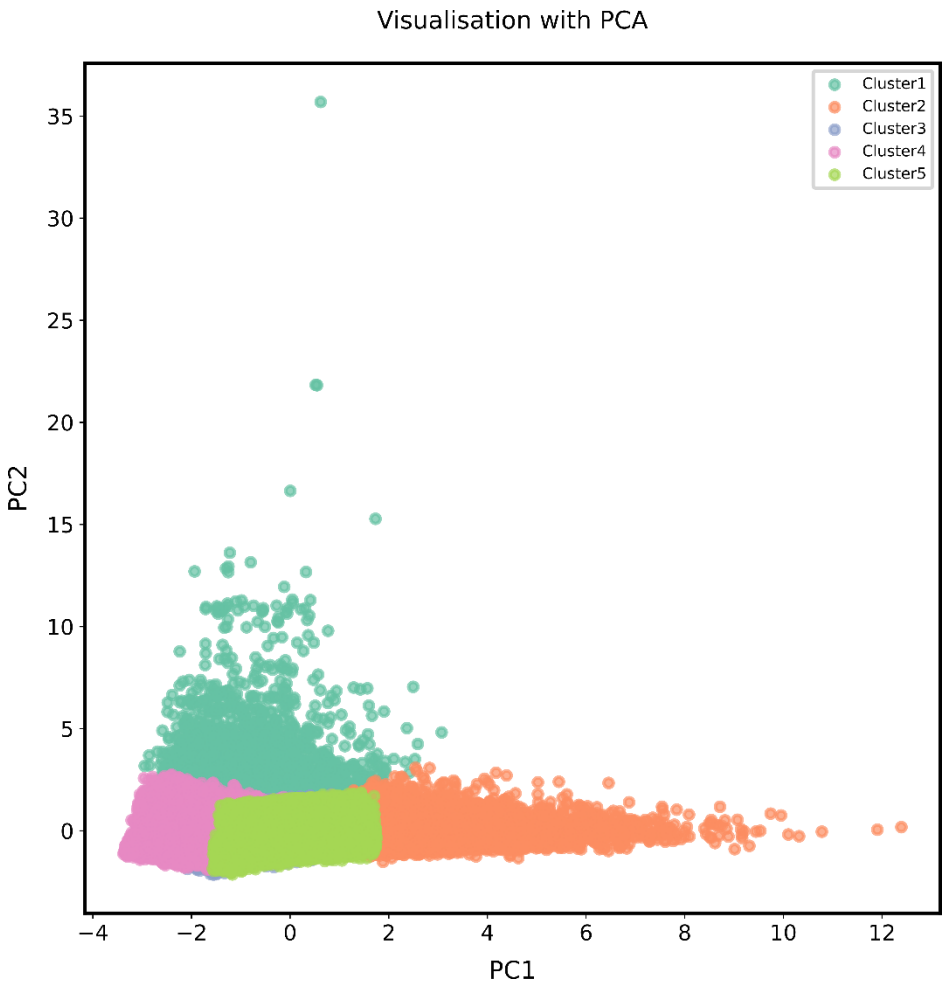


Figure 6: The resulting 5 clusters are projected onto the two principal axes , PC1 and PC2, that present with the most variance.

The t-SNE plot in Figure 7 produces a much higher resolution amongst clusters, albeit at a high computational cost. The clusters continue to overlap whereas the relatively small size of cluster 1 suggests segmentation with $k = 3$ or 4 could improve the picture.

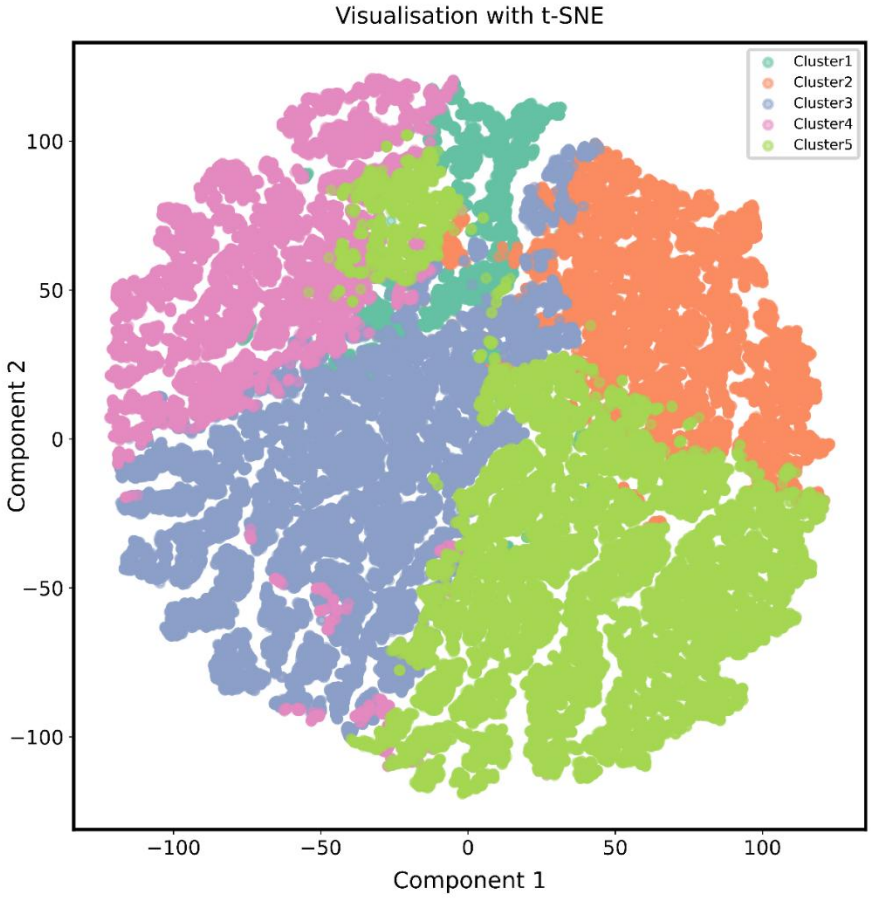


Figure 7: The t-SNE output following mapping of the 5-feature dataset onto 2D, using a perspective of 20.

Conclusions

The optimal number of clusters, k, has been determined for an e-commerce dataset, using both statistical approaches such as within-cluster variation and silhouette score analyses as well as a dendrogram interpretation of the resulting hierarchical structure. The three approaches combined indicated an optimal $k = 5$, which was used in the final segmentation of the dataset using k-means. The results were visualised in 2D following dimensionality reduction with PCA and t-SNE. The t-SNE method produced more interpretable and powerful results with better-resolved clusters. The t-SNE output suggested a reduced number of clusters such as $k = 3$ or 4 should be considered which was explored in Figure 8. While $k = 4$ appears optimal, the case of $k = 3$ would have resulted in a loss of nuance.

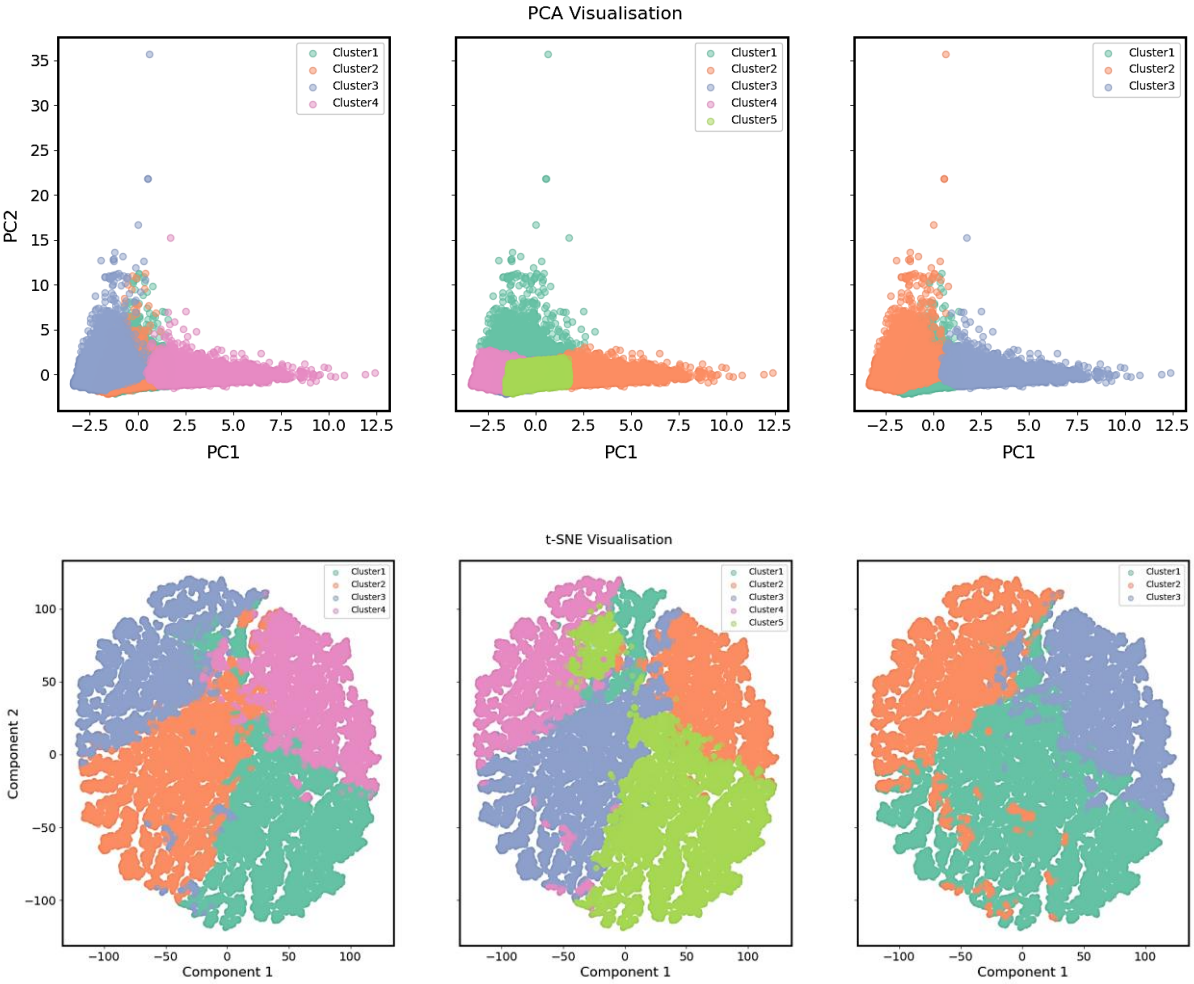


Figure 8: The effect of altering k on the PCA (top panel) and t-SNE outputs (bottom panel). From left-to-right, a scenario of $k = 3$, 5 and 4 has been visualised.